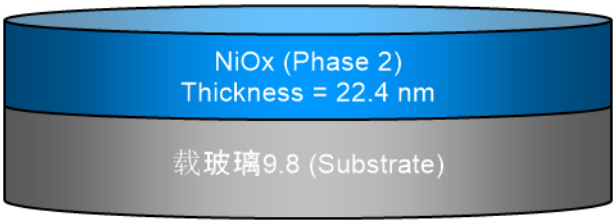
SEA regression report summary

Sample ID
5-1

Details	
Software and regression log	
Software about	Semilab - Spectroscopic Ellipsometry Analyzer - SEA
Software version	1.8.0.4
Officially licensed to	Linyang Jiangsu
Operator	operator
Date and time of regression	07-04-2026 16:59
Comments	

Layer structure	
Overview	

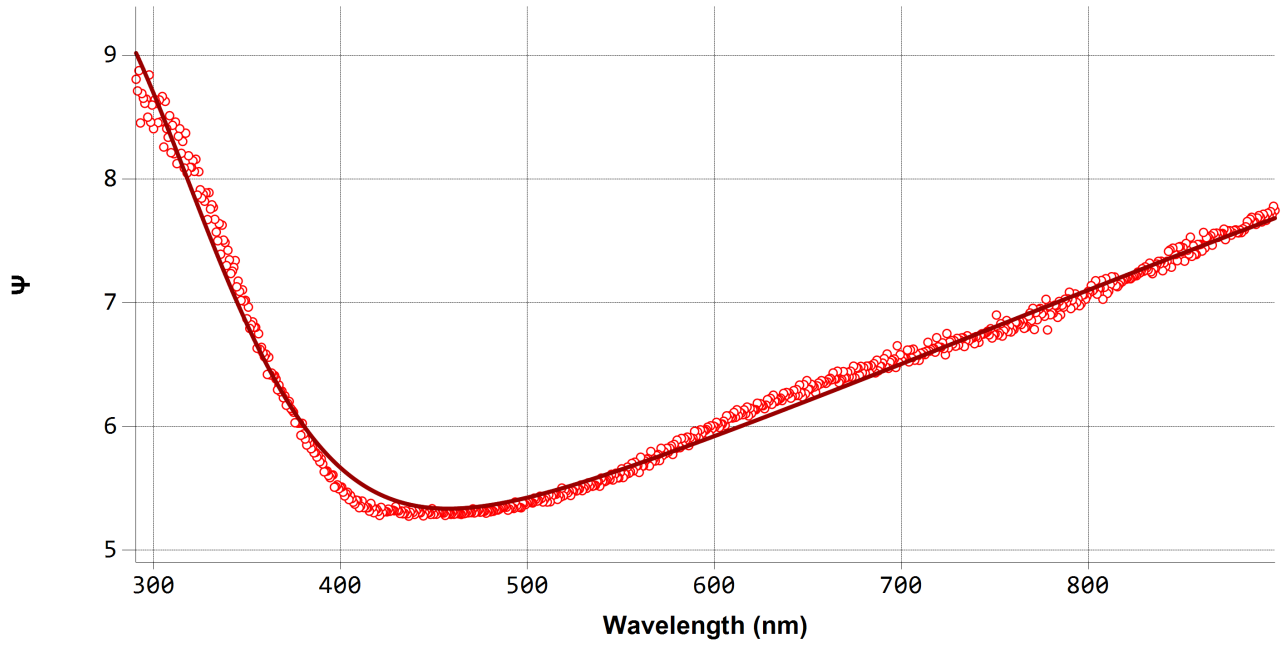


Optical model	
Phase 2	NiOx
Dispersion law	Tauc-Lorentz
	Gauss
	Lorentz

Regression results

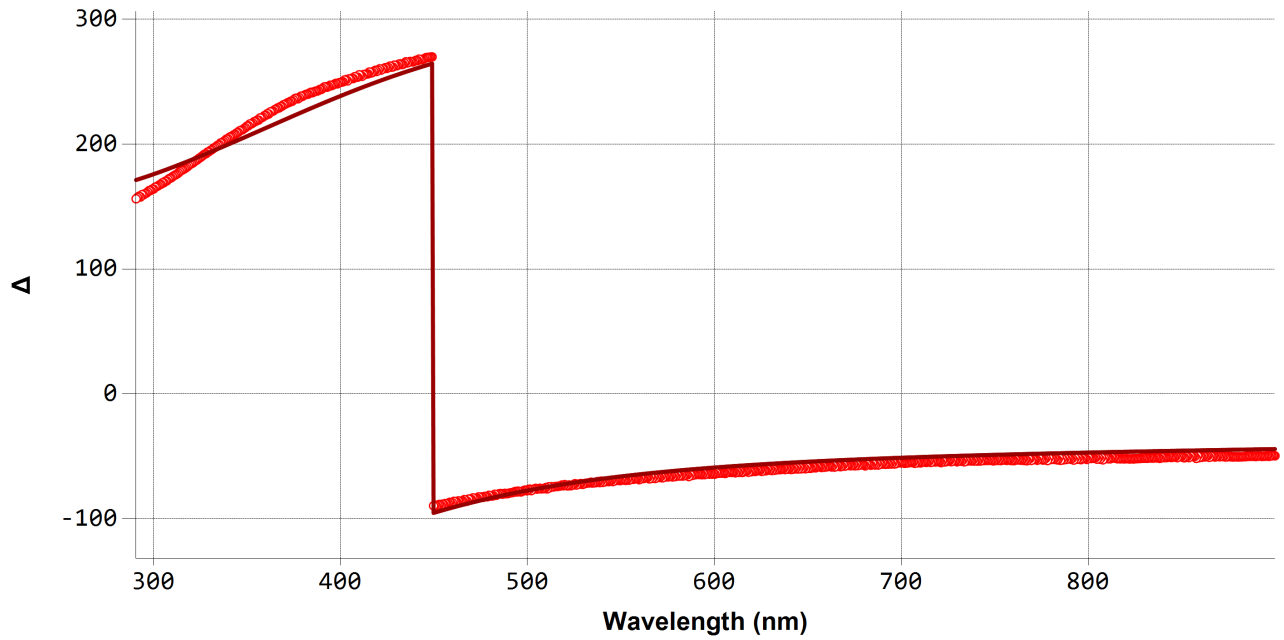
Measurement information				
Measurement file path	C:\Users\jijun.zhang\Desktop\SE2000□□\ □□\ 0813\NIO\MF-8KW-0.5-4.4-18\5-1.smdx			
Angle of Incidence	64.6°			
Regression details				
Regression 1 (EllipsoReflectance)				
Wavelength range	290.61 - 899.93 nm			
Angle of Incidence	64.6°			
Fit to	Ψ, Δ			
Angular Aperture	0°			
Fit algorithm	LMA			
Results				
Parameters	Value	Fitted	2 σ confidence limit	Unit
Model				
AOI Shift	0			°
Angular Aperture	0			°
Phase 2 (NiOx)				
Thickness	22.443	X	0.030307	nm
A (eV)	61.48773			eV
E0 (eV)	7.1813			eV
C (eV)	1.0016E-15			eV
Eg (eV)	2.95995			eV
Amp	1.68799	X	0.007089	
E0 (eV)	4.61996	X	0.019915	eV
Br (eV)	2.12908	X	0.036164	eV
f	1.71587			
E0 (eV)	2.42066			eV
Γ (eV)	5			eV
Eps_inf	1.4488E-05			
Derived parameters	Value			
Phase 2 (NiOx)				
n @ 632.8 nm	2.0476			
k @ 632.8 nm	0.2458			
Substrate (□□□9.8)				
n @ 632.8 nm	1.4817			
k @ 632.8 nm	0.0043			
Fit quality				
R^2	0.99432			
RMSE	0.07966			

Regression graphs



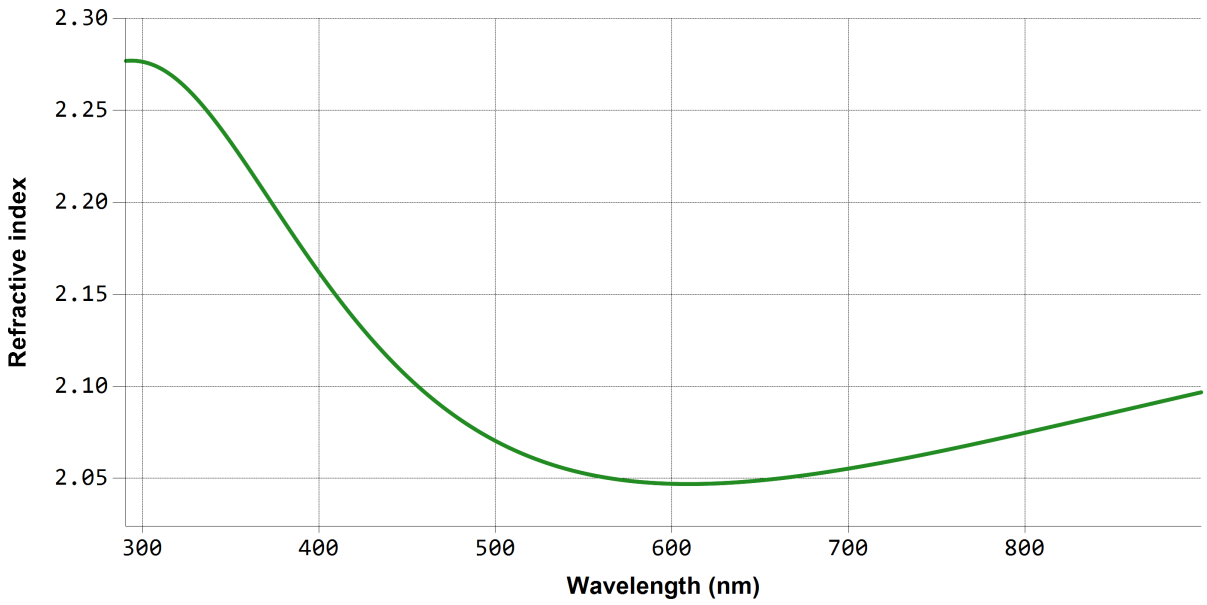
—	5-1 Measured	—	5-1 Fit

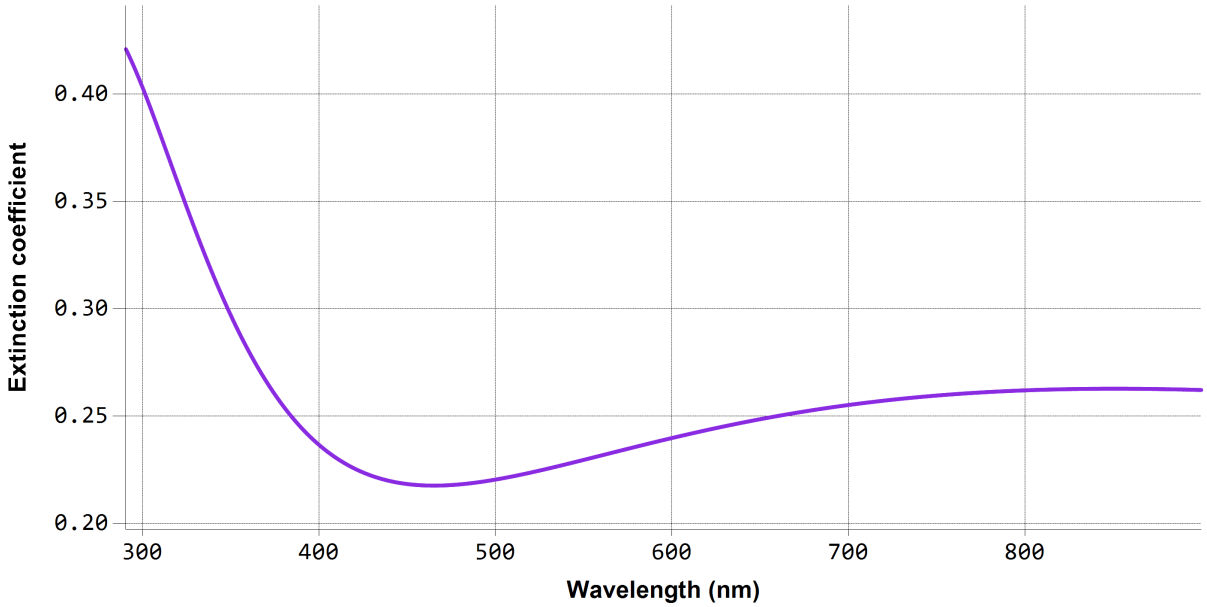
Regression graphs



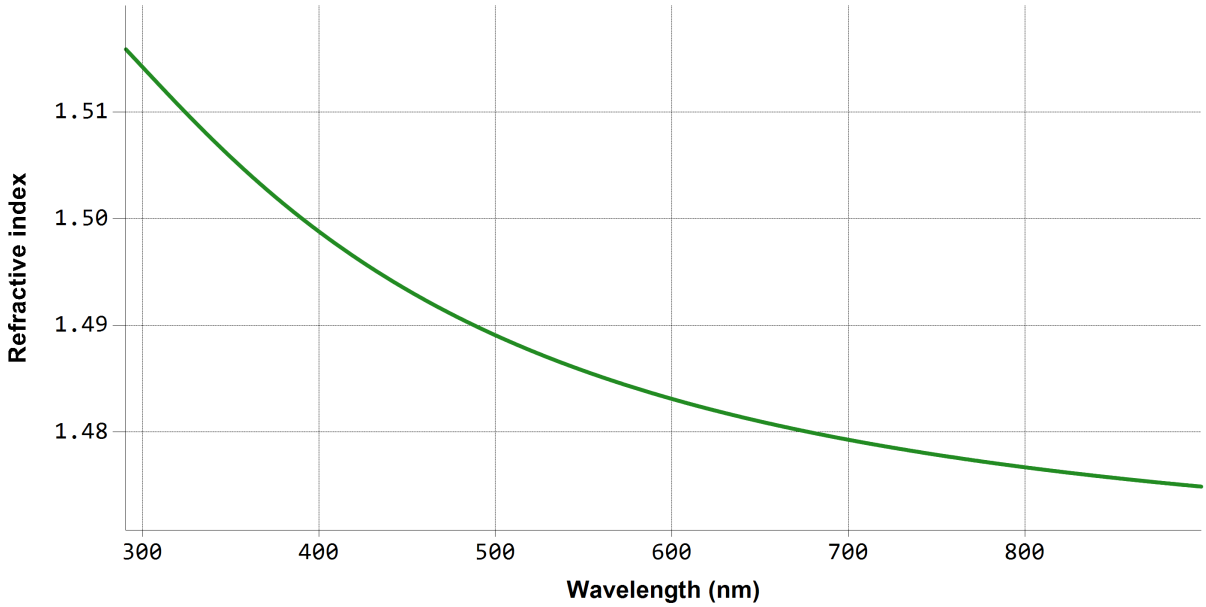
—	5-1 Measured	—	5-1 Fit

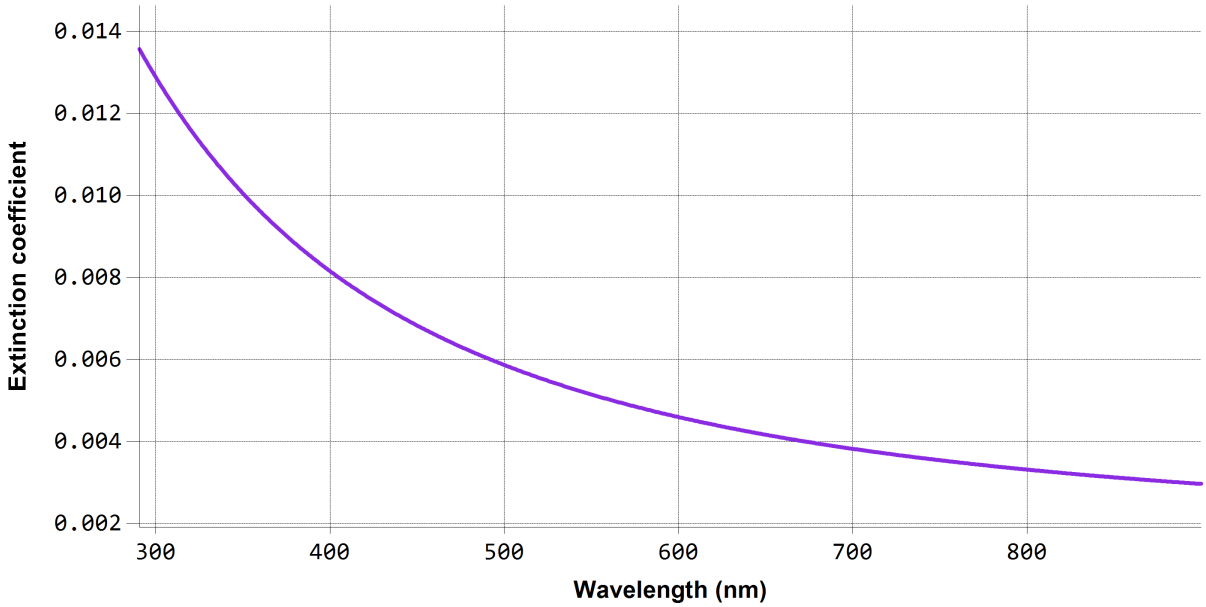
Phase 2 (NiOx) - Dispersion graphs





Substrate (□□□9.8) - Dispersion graphs





Correlation coefficients				
	Ph2 - NiOx - Thickness	Ph2 - Gauss[2] - Amp	Ph2 - Gauss[2] - E0 (eV)	Ph2 - Gauss[2] - Br (eV)
Ph2 - NiOx - Thickness	1	0.1309	-0.3962	-0.4884
Ph2 - Gauss[2] - Amp		1	-0.7865	-0.4433
Ph2 - Gauss[2] - E0 (eV)			1	0.7807
Ph2 - Gauss[2] - Br (eV)				1